

GENE REGULATORY NETWORK INFERENCE FROM INTEGRATIVE MULTI-OMICS DATA

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Weekly Update

A Recap

- The goal: figure out which transcription factors (TFs) switch which genes on or off in a cell, this map is called a Gene Regulatory Network (GRN).
- The database we use (SC-MO-GRN-DB) is special: it doesn't give us one plain list of "true" edges. It labels every edge with how confident we should be about it.
- Three confidence levels:
 - Localization (from ChIP-seq): the TF is physically seen sitting near the gene. Lots of these edges, but sitting nearby doesn't always mean it's actually regulating the gene.
 - Perturbation (from knockout/CRISPR experiments): when the TF is removed, the gene's expression actually changes. Fewer edges, but this is real functional proof.
 - Dual-evidence: edges that have both the smallest set, but the most trustworthy one we have.

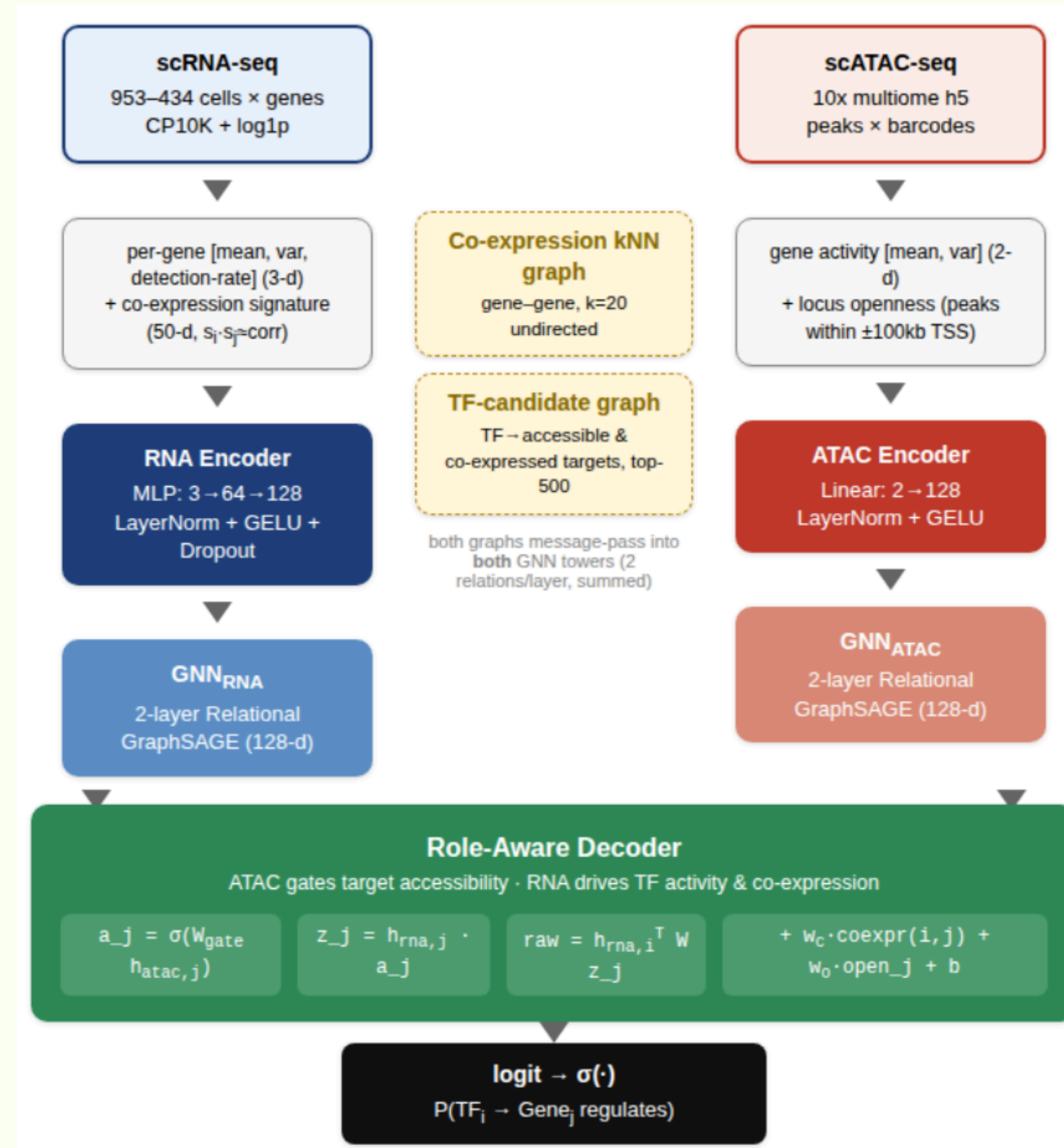
Baseline Implementation

- Added a strong new baseline: implemented and trained scMultiomeGRN, a recently published multi-omic GNN (4.08 million parameters), on the exact same K562 data so we can compare fairly.
- Simplified the training plan: the model now trains on only two stages - localization, then perturbation. The dual-evidence tier is never trained on; it is used purely as a final, unseen test.
- Made the data splitting rock-solid: built a single, global way of splitting edges into train/validation/test so that no edge can ever be "seen" in one tier and "tested" in another, even though the tiers overlap.
- Ran a full set of experiments (10 variants) to check which parts of the model and training plan actually matter.
- Found and measured an honest weakness: training on perturbation after localization makes the model forget what it learned about localization. We measured exactly how bad this is and how much a fix (memory replay) helps.

Updated Architecture

- Idea: gene expression (RNA) and DNA accessibility (ATAC) are not the same kind of signal, so we stop mixing them together early. A gene can only be regulated if (a) its DNA region is open/accessible, and (b) it is co-expressed with the TF, these are two separate checks, not one blended score.
- RNA side: a small neural network turns each gene's expression pattern (mean, variability, how often it's detected, plus a 50-number "co-expression fingerprint") into an embedding.
- ATAC side: a similarly small network turns each gene's accessibility pattern into its own embedding.
- Both sides talk to their gene neighbors using a graph neural network, so each gene also learns from nearby co-expressed genes and candidate TF targets.
- The two sides only meet at the very end, in a "Role-Aware Decoder": the accessibility embedding acts like a gate that decides how much of the RNA signal is allowed through, before the final TF→gene score is computed.
- Total size: 306,691 parameters which is about 13× smaller than the new scMultiomeGRN baseline (4,077,914 parameters).

Updated Architecture



About the dataset

- Cell line used: K562 (human), the main cell type with all three evidence tiers available.
- What we have: 953 cells of RNA data, ~1,126 cells of ATAC data, and a final list of 22,943 genes, 225 of which are TFs.
- Edge counts after cleanup: Localization $\approx$ 1.17M edges, Perturbation $\approx$ 230K edges, Dual-evidence $\approx$ 27K edges.
- The important catch: these tiers are not separate, independent lists, they overlap heavily. At K562, dual-evidence is 100% contained inside perturbation, and 77.5% contained inside localization.
- Why this matters: if we trained the model directly on dual-evidence, we'd mostly be re-teaching it things it already saw in an earlier stage, that wouldn't really test whether it generalizes, just whether it memorizes. This is exactly why we hold dual-evidence out as a pure test set instead

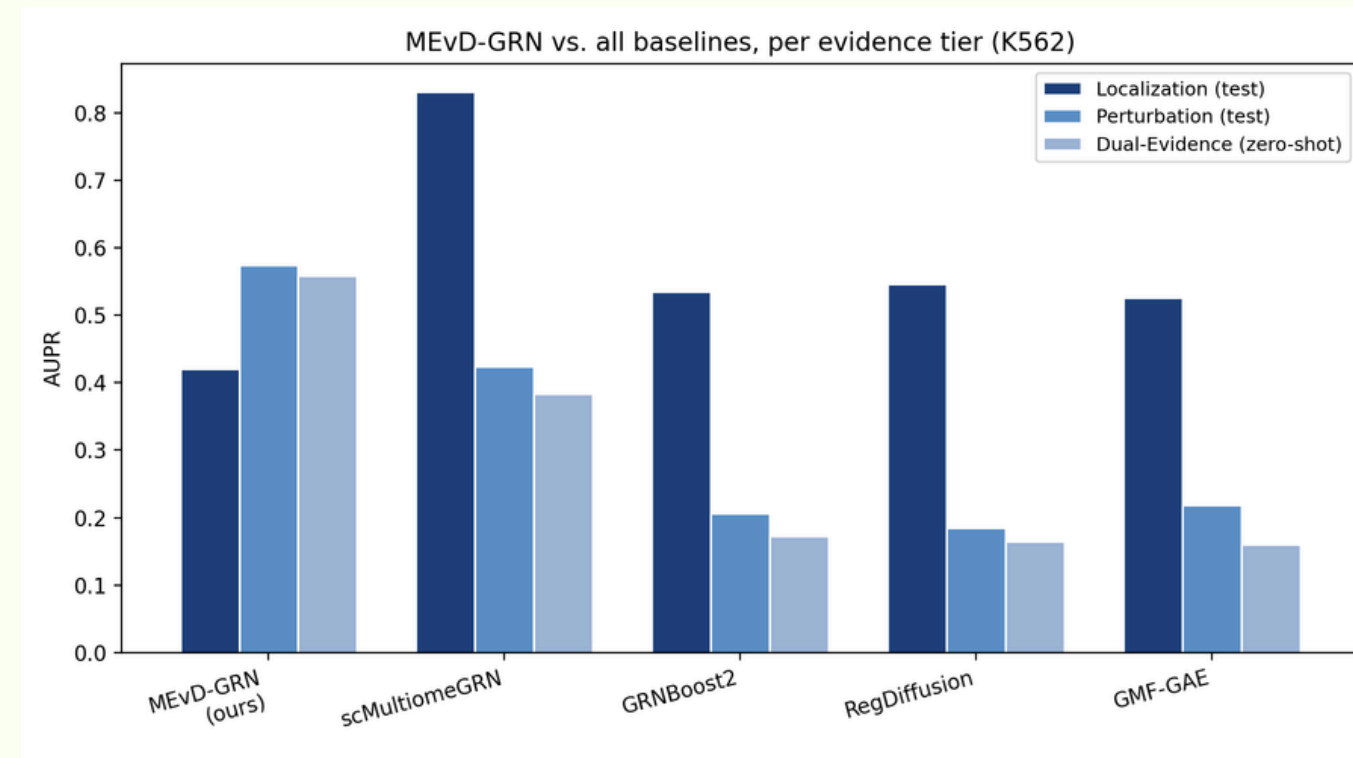
Data Preprocessing

- **RNA**: library-size normalization followed by $\log_1 p$ transform ($\log(1+\text{count})$)
- **ATAC**: dataset-specific normalization with log-transform applied in the dataloader
- **Gene-peak candidate sets** (C_g): for each gene, candidate peaks selected within a $\pm 5\text{kb}$ window around the gene, ranked by genomic distance, and truncated to a fixed maximum width (capped at 50 peaks/gene) for batching which avoids fixed/hard-coded peak-to-gene mapping
- **Bulk-derived priors**: regulator–target links mapped into the model's gene vocabulary; matched edges used as noisy positive labels in Stage 2; all other pairs treated as unlabeled
- QC applied to remove low-quality cells and low-coverage genes/peaks
- Barcode alignment performed across paired RNA/ATAC modalities (already paired by construction in multiome assays)

Updating Training Paradigm

- **Stage 1 - Localization:** the model first learns the broad picture of which TFs can reach which genes (30 training passes).
- **Stage 2 - Perturbation:** the model is then fine-tuned to tell "just binds" apart from "actually regulates" (15 more training passes).
- **Dual-evidence** - never trained on: it is used only once, at the very end, purely to test the model.
- **Splitting rule:** every unique TF-gene pair is assigned to train/validation/test once, globally, before being divided up per tier. This guarantees a pair can never be "training data" in one tier and "test data" in another, so the zero-shot test on dual-evidence is genuinely fair.
- half of the "negative" examples shown to the model are deliberately tricky ones (pairs that bind but don't regulate), to sharpen the model's judgement.

Results



Method	Localization AUPR	Perturbation AUPR	Dual-evidence AUPR
MEvD-GRN (ours)	0.420	0.575	0.558
scMultiomeGRN	0.831	0.423	0.384
GRNBoost2	0.535	0.206	0.173
RegDiffusion	0.546	0.185	0.165
GMF-GAE	0.526	0.218	0.160

Ablation Study

What we removed/changed	Dual-evidence AUPR	What it tells us
Train both tiers together (no strict order)	0.788	Beats our sequential plan — sequencing alone isn't the win we expected
Add a consolidation stage + replay	0.940	Best score overall — the strongest fix for forgetting
Remove the graph neural network	0.241	Worst score — the graph part is the most important piece
Remove ATAC (RNA only)	0.543	Barely changes — ATAC isn't adding much yet, worth digging into
Change the fusion style (gated / concat)	0.546 / 0.555	Barely changes — a smaller effect than expected



THANK YOU!

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